

Software Requirements Specification

For

FoodBridge : Real-Time Food Donation & Distribution Platform

Version : 1

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Revision History

Name	Date	Reason for changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of FoodBridge, a web-based food donation and distribution platform. FoodBridge connects restaurants with surplus food, NGOs, volunteers, donors, and administrators through a single platform. This document serves as a blueprint for the design, development, testing, deployment, and maintenance of the system, ensuring that all stakeholders share a common understanding of the project requirements.

1.2 Intended Audience and Reading Suggestions

This document is intended for project developers, testers, project managers, clients, faculty members, and end users of the FoodBridge system. Developers should use this document as a technical guide during implementation, testers should use it for validating system functionality, project managers should use it for planning and monitoring progress, and stakeholders should use it to understand the overall capabilities, objectives, and limitations of the system.

1.3 Project Scope

FoodBridge is a real-time food donation management platform developed to reduce food wastage and improve food accessibility for people in need. Restaurants can publish details of surplus food, NGOs can claim available donations, volunteers can deliver food, and donors can contribute financially to support NGO activities. The system provides secure user authentication, donation management, volunteer coordination, delivery tracking, notifications, reports, dashboards, and an administrative module for complete system management.

- Restaurant Registration and Login
- NGO Registration and Login
- Volunteer Registration and Login
- Donor Registration and Login
- Food Donation Management
- Donation Claiming
- Volunteer Assignment
- Delivery Tracking
- Monetary Donations
- Real-time Notifications
- Dashboard and Reports
- Admin Management

1.4 References

- IEEE 830 Software Requirements Specification Standard
- Spring Boot Documentation
- React Documentation
- MySQL Documentation
- Oracle Java Documentation
- JWT Documentation
- REST API Design Guidelines
- AWS Cloud Documentation
- UML 2.5 Specification

2. Overall Description

2.1 Product Perspective

FoodBridge is a web-based platform designed to bridge the gap between food donors and organizations that distribute food to people in need. The system provides a centralized platform where restaurants can donate surplus food, NGOs can claim available donations, volunteers can manage food pickup and delivery, and donors can contribute financially to support charitable activities. The platform improves transparency, minimizes food wastage, and ensures efficient coordination among all stakeholders through a secure and user-friendly interface.

2.2 Product Features

- User Registration and Secure Login
- Restaurant Profile Management
- NGO Profile Management
- Volunteer Management
- Donor Registration
- Food Donation Posting
- Donation Claim Request
- Donation Approval and Rejection
- Volunteer Assignment
- Pickup and Delivery Tracking
- Real-time Notifications
- Online Monetary Donations
- Dashboard with Analytics
- Report Generation
- User Profile Management
- Role-based Access Control
- Admin Panel

2.3 User Classes and Charecteristics

Restaurant

Restaurants register on the platform and post details of surplus food available for donation. They can monitor donation requests, approve NGOs, and track donation status.

NGO

NGOs browse available food donations, claim suitable donations, coordinate with volunteers, and ensure food reaches beneficiaries.

Volunteer

Volunteers accept assigned deliveries, collect food from restaurants, deliver it to NGOs, and update delivery status.

Donor

Donors contribute monetary donations to NGOs through the platform and can track their donation history.

Administrator

Administrators manage users, verify NGOs, monitor donations, generate reports, and maintain overall system security.

2.4 Operating Environment

Hardware

- Computer with minimum Intel Core i3 / Ryzen 3 Processor
- 8 GB RAM or higher
- Stable Internet Connection

Software

- Windows 10/11 or macOS
- Visual Studio Code
- Spring Boot
- React.js
- MySQL Server
- MySQL Workbench
- Java JDK 21
- Git
- Postman
- Docker (Optional)

Deployment

- AWS Cloud
- Apache Tomcat (Embedded in Spring Boot)

2.5 Design and Implementation Constraints

- Internet connection is mandatory.
- English is the only supported language.
- JWT authentication is used for security.
- MySQL is used as the primary database.
- Users must register before accessing services.
- Food donation records cannot be permanently deleted by normal users.
- The system follows REST API architecture.

2.6 User Documentation

FoodBridge provides a simple and user-friendly interface with clear navigation menus, dashboard pages, validation messages, and help sections. User guides and system documentation will assist restaurants, NGOs, volunteers, donors, and administrators in using the application effectively.

2.7 Assumptions and Dependencies

Assumptions

- Users have internet connectivity.
- Restaurants provide accurate food information.
- NGOs are verified before using the system.
- Volunteers are available for delivery.
- Donors use secure online payment methods.

Dependencies

- Spring Boot Framework
- React.js
- MySQL Database
- Java JDK
- JWT Authentication
- AWS Cloud Services
- Email Notification Service

3. System Features

3.1 User Login

3.1.1 Description and Priority

- **Description:** All users (Restaurant, NGO, Volunteer, Donor, and Administrator) can securely log in using their registered email address and password.
- **Priority:** High

3.1.2 Stimulus/Response Sequences

The user opens the FoodBridge application, enters a valid email address and password, and clicks the **Login** button. If the credentials are correct, the system authenticates the user and redirects them to their respective dashboard based on their role. If the credentials are invalid, an appropriate error message is displayed.

3.1.3 Functional Requirements

- The system shall authenticate users using email and password.
- Passwords shall be stored in encrypted form.
- JWT tokens shall be generated after successful login.
- Unauthorized users shall not access protected pages.
- The system shall automatically expire inactive sessions.

3.2 Dashboard

3.2.1 Description and Priority

- **Description:** Each authenticated user is provided with a personalized dashboard displaying information relevant to their role.
- **Priority:** High

3.2.2 Stimulus/Response Sequences

After successful login, the system retrieves user-specific information and displays the dashboard. Restaurants see their donations, NGOs see claimed donations, volunteers see assigned deliveries, donors see donation history, and administrators see overall statistics.

3.2.3 Functional Requirements

- Display role-specific dashboard.
- Display recent activities.
- Display notifications.
- Display donation statistics.

- Allow quick navigation to system modules.

3.3 Food Donation Management

3.3.1 Description and Priority

- **Description:** Restaurants can create, edit, update, and remove surplus food donation listings.
- **Priority:** High

3.3.2 Stimulus/Response Sequences

A restaurant logs into the system, opens the donation page, enters food details, pickup location, expiry time, quantity, and submits the donation. The system validates the information and publishes the donation for NGOs.

3.3.3 Functional Requirements

- Add food donation.
- Edit donation details.
- Delete donation before claim.
- Upload food images.
- Set food expiry time.
- Notify nearby NGOs.

3.4 Donation Claim Management

3.4.1 Description and Priority

- **Description:** NGOs can browse available donations and submit claim requests.
- **Priority:** High

3.4.2 Stimulus/Response Sequences

The NGO selects a food donation and clicks **Claim Donation**. The restaurant receives the request and either approves or rejects it. The NGO is notified of the decision.

3.4.3 Functional Requirements

- View available donations.
- Claim donation.
- Cancel claim request.
- Receive approval/rejection notification.
- View claim history.

3.5 Volunteer Management

3.5.1 Description and Priority

- **Description:** Volunteers manage food pickup and delivery.
- **Priority:** High

3.5.2 Stimulus/Response Sequences

The volunteer receives a delivery assignment, accepts it, collects food from the restaurant, delivers it to the NGO, and updates the delivery status.

3.5.3 Functional Requirements

- View assigned deliveries.
- Accept delivery.
- Update delivery status.
- Mark delivery completed.
- View delivery history.

3.6 Donor Management

3.6.1 Description and Priority

- **Description:** Donors can financially support NGOs using secure online payments.
- **Priority:** Medium

3.6.2 Stimulus/Response Sequences

The donor selects an NGO, enters the donation amount, completes payment through the payment gateway, and receives a confirmation receipt.

3.6.3 Functional Requirements

- View NGO list.
- Donate online.
- View payment history.

- Download donation receipts.
- Receive payment confirmation.

3.7 Admin Module

3.7.1 Description and Priority

- **Description:** The administrator manages all users, donations, reports, and system settings.
- **Priority:** High

3.7.2 Stimulus/Response Sequences

The administrator logs into the admin dashboard and performs tasks such as verifying NGOs, managing users, monitoring donations, and generating reports.

3.7.3 Functional Requirements

- Manage all users.
- Verify NGOs.
- View all donations.
- Generate reports.
- Block suspicious users.
- Monitor platform activity.

3.8 Notification System

3.8.1 Description and Priority

- **Description:** The system sends notifications for important activities.
- **Priority:** Medium

3.8.2 Stimulus/Response Sequences

Whenever a donation is posted, claimed, approved, delivered, or a payment is completed, the system sends notifications to the relevant users.

3.8.3 Functional Requirements

- Notify restaurants of claim requests.
- Notify NGOs of approvals.
- Notify volunteers of new assignments.
- Notify donors after successful payments.
- Notify administrators of important system events.

4. External Interface Requirements

4.1 User Interfaces

FoodBridge provides a clean, responsive, and user-friendly web interface accessible through modern web browsers. Each user role is presented with a dedicated dashboard containing only the features relevant to their responsibilities. The interface includes registration and login pages, restaurant dashboard, NGO dashboard, volunteer dashboard, donor dashboard, and administrator dashboard. The application follows responsive web design principles to ensure usability on desktops, laptops, tablets, and mobile devices.

4.2 Hardware Interfaces

Server Side

- Cloud server hosting the backend application.
- Minimum 8 GB RAM and multi-core processor.
- Internet connectivity for continuous operation.

Client Side

- Desktop or Laptop Computer
- Smartphone or Tablet
- Keyboard and Mouse/Touchscreen
- Stable Internet Connection

4.3 Software Interfaces

Backend

- Spring Boot Framework
- Java JDK 21
- Spring Security
- JWT Authentication
- Hibernate ORM

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

- Tailwind CSS

Database

- MySQL Server
- MySQL Workbench

Development Tools

- Visual Studio Code
- IntelliJ IDEA / Spring Tool Suite (STS)
- Git
- Postman
- Maven

4.4 Communication Interfaces

FoodBridge communicates between the client and server using RESTful APIs over the HTTPS protocol. Data is exchanged in JSON format. JWT-based authentication is used to secure API requests. Email notifications are sent using SMTP services, and cloud deployment is performed using AWS infrastructure.

5. Other Non-Functional Requirements

5.1 Performance Requirements

FoodBridge shall provide fast and reliable performance under normal operating conditions. The system should load dashboard pages within three seconds under normal network conditions. It should support multiple users accessing the application simultaneously without significant degradation in performance. The application should efficiently process food donation requests, volunteer assignments, and payment transactions while maintaining data consistency. The system should be scalable to accommodate future growth in the number of users and donations.

5.2 Safety Requirements

FoodBridge shall ensure that food donation information is accurate and accessible only to authorized users. The application shall validate all user inputs to reduce errors and prevent incorrect data entry. Backup and recovery mechanisms shall be available to minimize data loss in case of system failure. Error messages should clearly guide users without exposing sensitive system information.

5.3 Security Requirements

The FoodBridge system shall implement strong security mechanisms to protect user data and application resources. The security requirements include:

- User authentication using email and password.
- JWT-based authorization for secure API access.
- Passwords shall be stored using encrypted hashing algorithms.
- All communication shall occur over HTTPS.
- Role-based access control for Restaurants, NGOs, Volunteers, Donors, and Administrators.
- Input validation to prevent SQL Injection and Cross-Site Scripting (XSS).
- Secure database access with proper permission management.
- Session timeout for inactive users.
- Regular database backup and recovery procedures.

5.4 Software Quality Attributes

The FoodBridge application is designed to satisfy the following software quality attributes:

- **Usability:** The interface shall be simple, intuitive, and easy to navigate for all users.
- **Reliability:** The system shall provide consistent and accurate operation with minimal downtime.

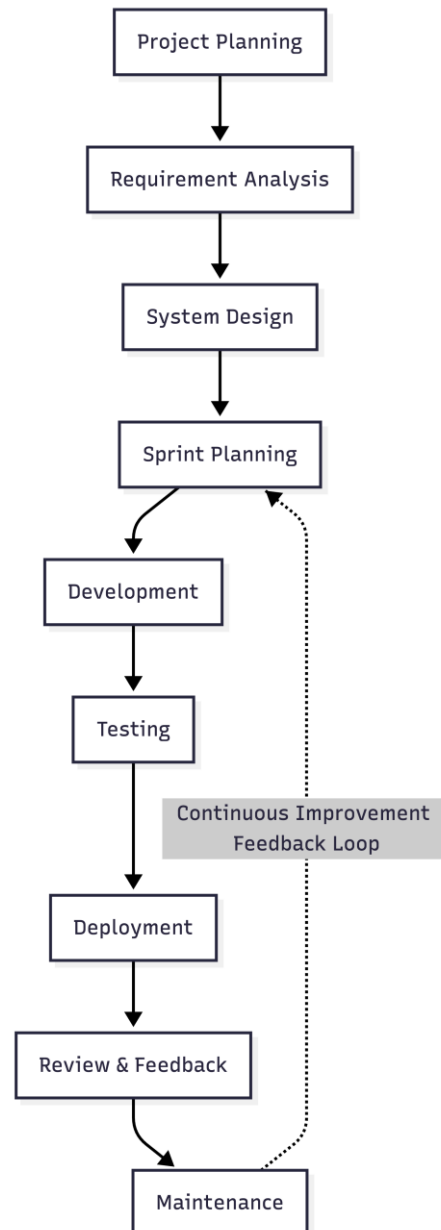
- **Availability:** The application should be available 24×7 except during scheduled maintenance.
- **Performance:** The application shall respond quickly under normal workloads.
- **Scalability:** The system shall support increasing numbers of users and food donations.
- **Maintainability:** The codebase shall be modular, documented, and easy to maintain.
- **Portability:** The application shall run on modern web browsers and different operating systems.
- **Security:** Sensitive information shall be protected using industry-standard security practices.
- **Interoperability:** The system shall communicate with external services such as payment gateways, email services, and cloud platforms.

Appendix A: Glossary

Sr No	Abbreviation	Full Form
1	API	Application Programming Interface
2	AWS	Amazon Web Services
3	DBMS	Database Management System
4	HTML	Hypertext Markup Language
5	HTTP	Hypertext Transfer Protocol
6	HTTPS	Hypertext Transfer Protocol Secure
7	JWT	Json Web Token
8	NGO	Non-Governmental Organization
9	ORM	Object Relational Mapping
10	REST	Representational State Transfer
11	SQL	Structured Query Language
12	SRS	Software Requirements Specification
13	UI	User Interface
14	UML	Unified Modeling Language
15	URL	Unified Resource Locator

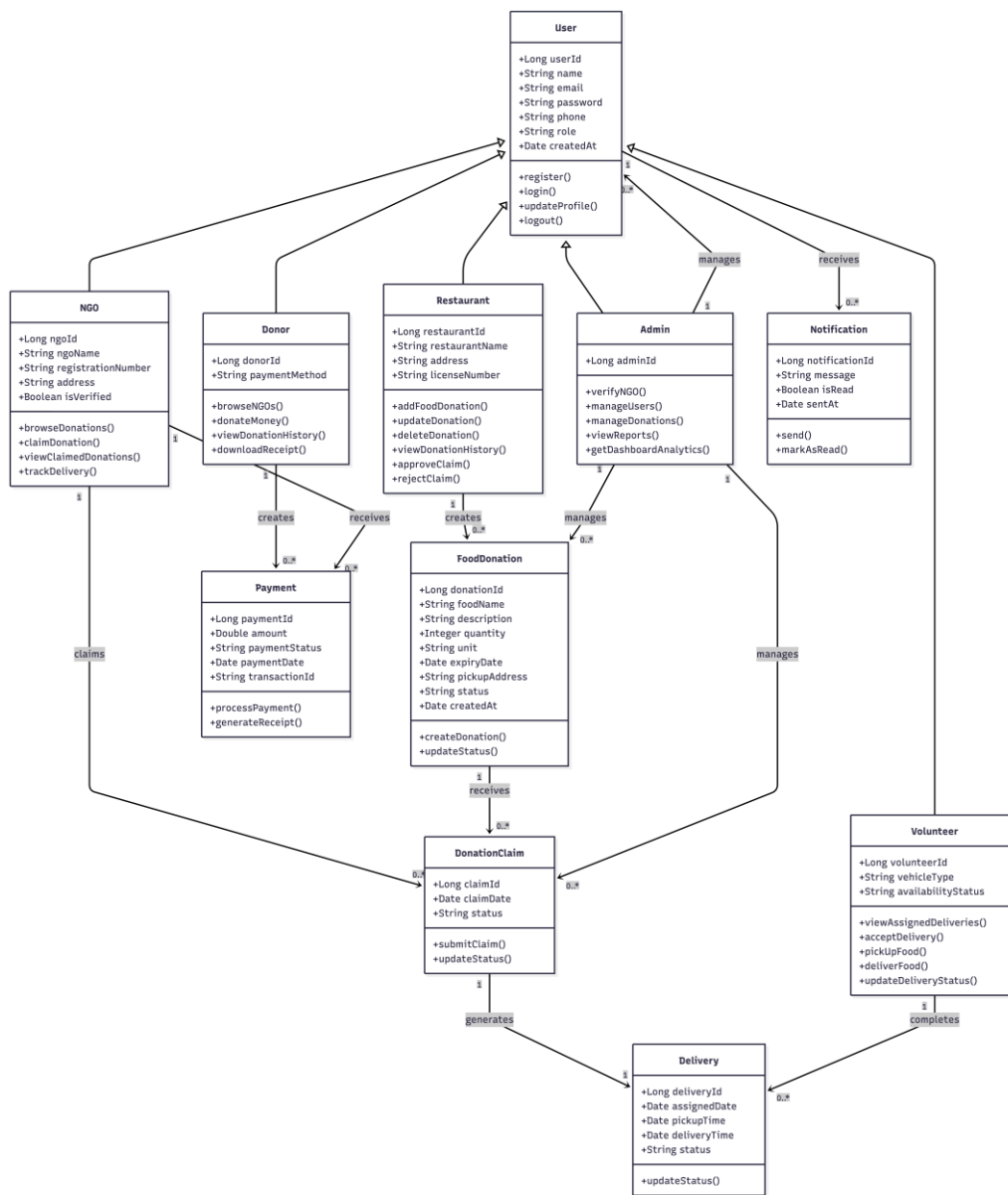
Appendix B: Analysis Models

- **Software Development Approach**

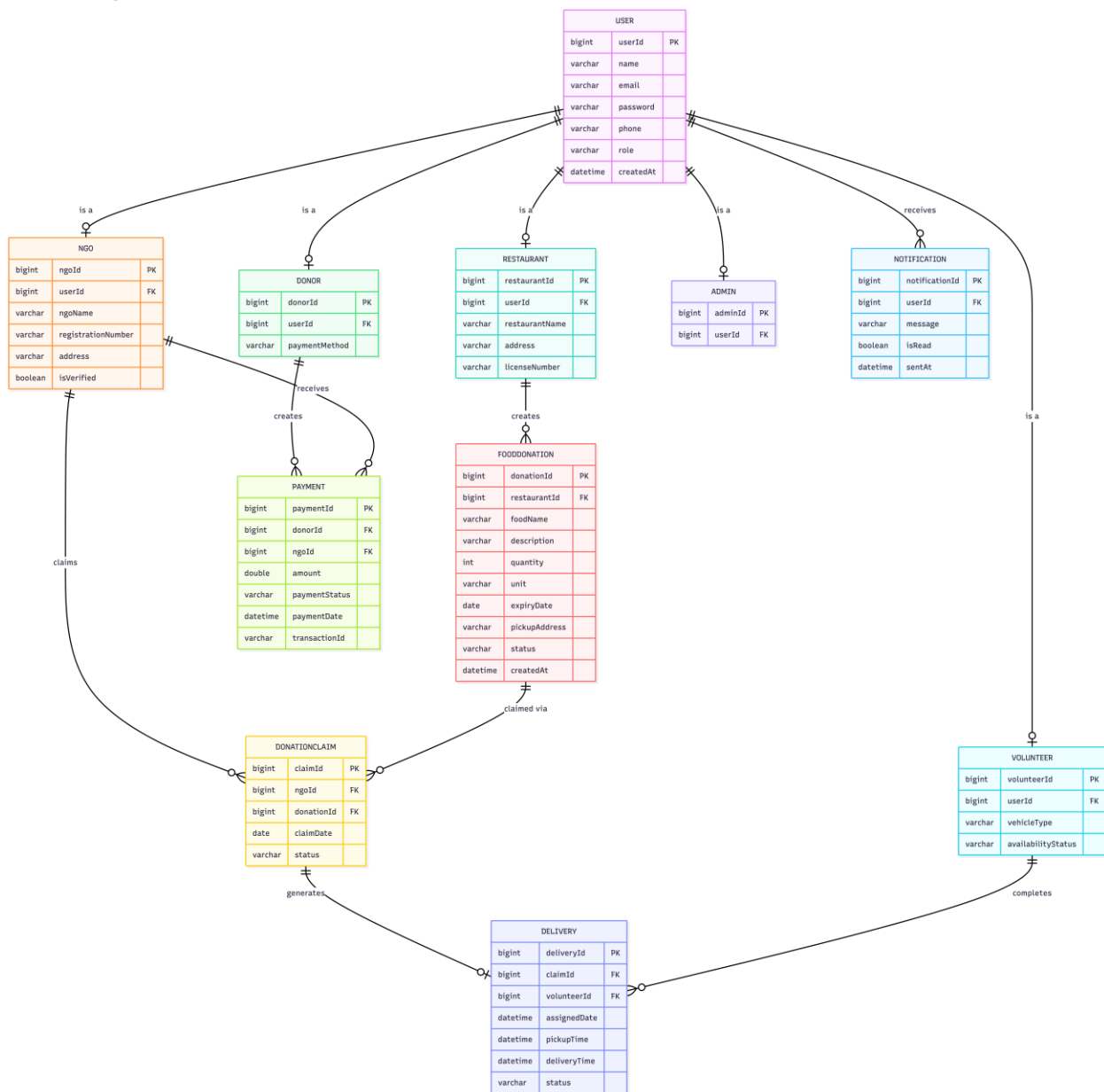


• Class

Diagram



• ER Diagram



- Data Flow Diagram**

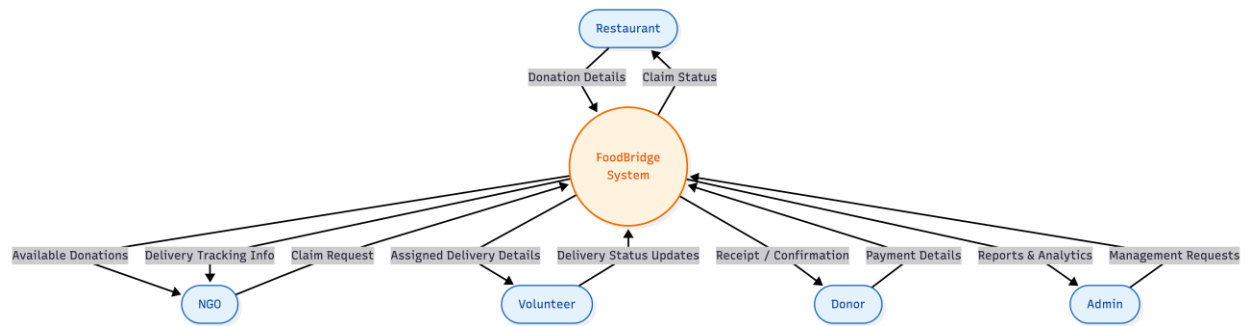


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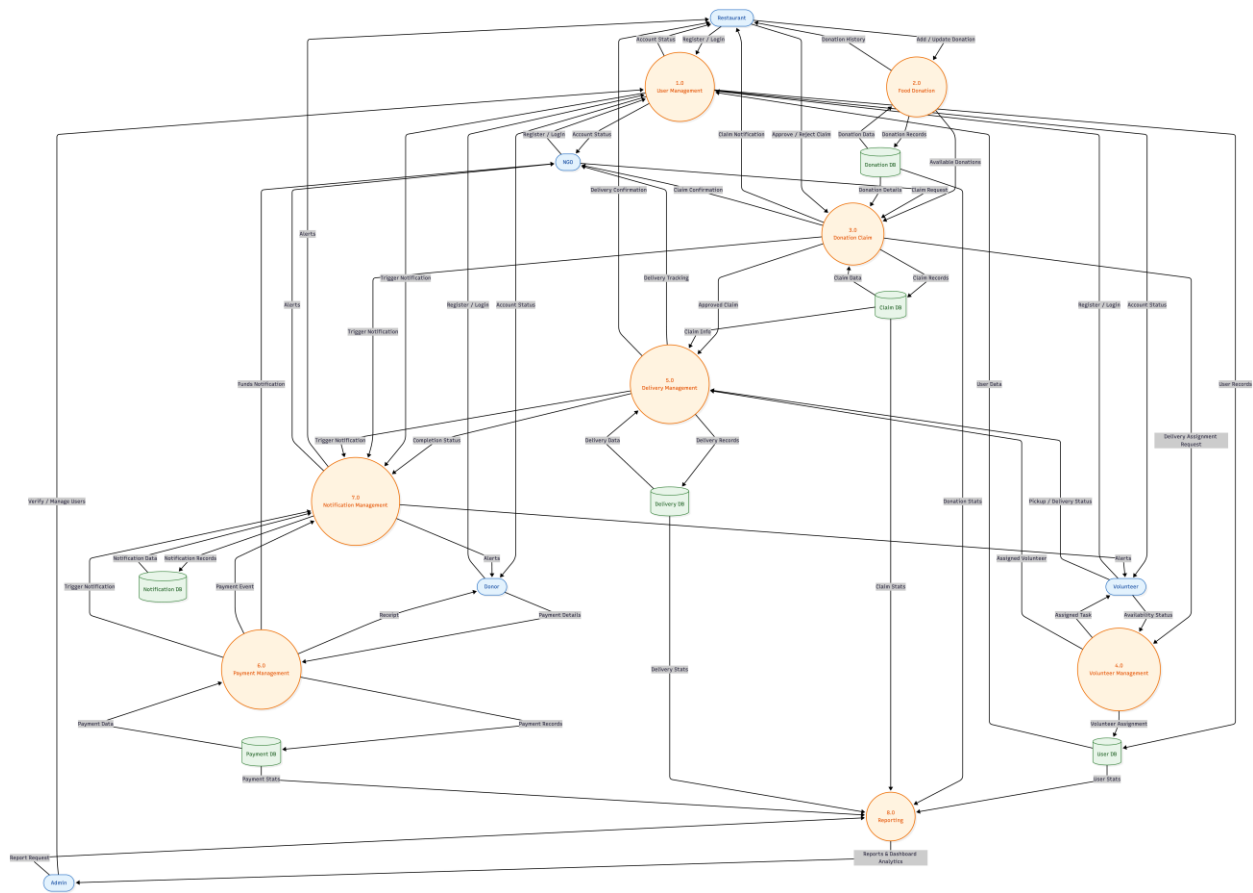
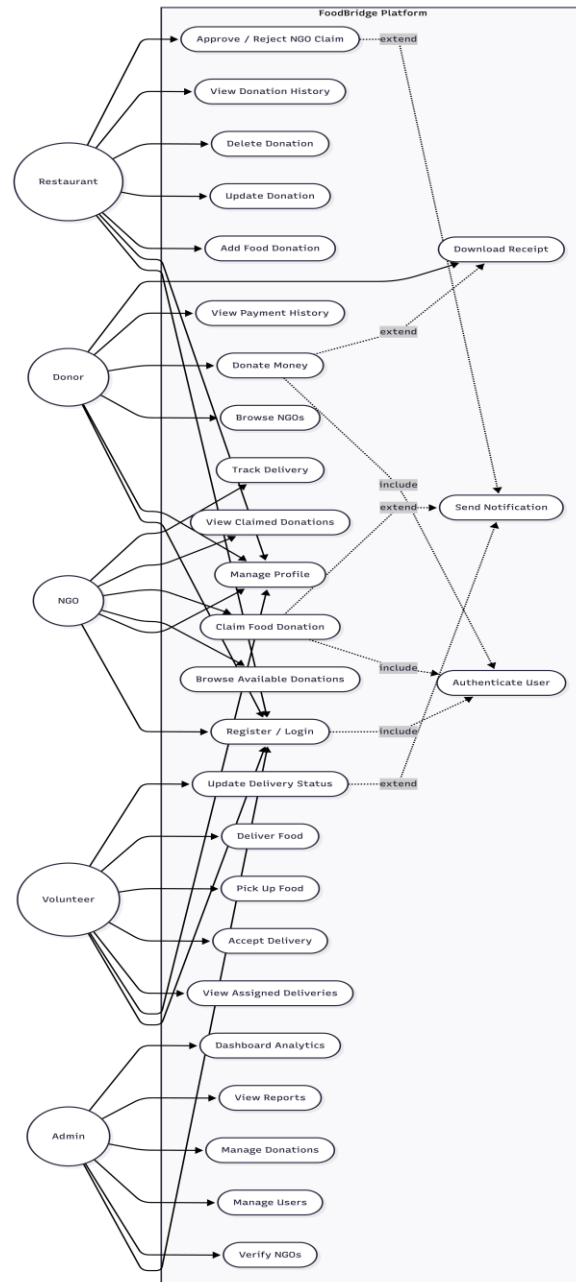
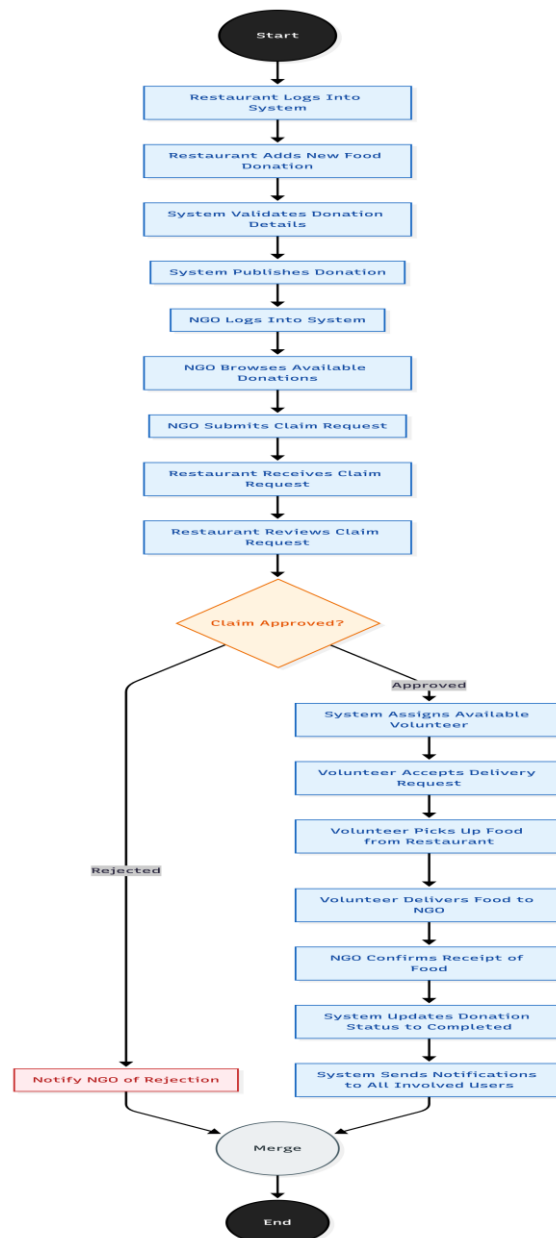


Fig : DFD1

- Use case Diagram**



- Activity Diagram



• Sequence Diagram

